torch.functional.unique_consecutive#

https://pytorch.org/docs/stable/generated/torch.functional.unique_consecutiv

Description:

Eliminates all but the first element from every consecutive group of equivalent elements. This function is different from torch.unique() in the sense that this function only eliminates consecutive duplicate values. This semantics is similar to std::unique in C++. input (Tensor) – the input tensor return_inverse (bool) – Whether to also return the indices for where elements in the original input ended up in the returned unique list.

Function Signature

```
torch.functional.unique_consecutive(*args, **kwargs)
[source]#

Parameters

Returns

Return type
```

Returns:

(Tensor, Tensor (optional), Tensor (optional))

Code Examples

```
>>> x = torch.tensor([1, 1, 2, 2, 3, 1, 1, 2])
>>> output = torch.unique_consecutive(x)
>>> output
tensor([1, 2, 3, 1, 2])

>>> output, inverse_indices = torch.unique_consecutive(x,
return_inverse=True)
>>> output
tensor([1, 2, 3, 1, 2])
>>> inverse_indices
tensor([0, 0, 1, 1, 2, 3, 3, 4])

>>> output, counts = torch.unique_consecutive(x,
return_counts=True)
>>> output
tensor([1, 2, 3, 1, 2])
>>> counts
tensor([2, 2, 1, 2, 1])
```

Notes & Warnings

> **NOTE** Note This function is different from torch.unique() in the sense that this function only eliminates consecutive duplicate values. This semantics is similar to std::unique in C++.

Detailed Documentation

Documentation

Eliminates all but the first element from every consecutive group of equivalent elements.

This function is different from torch.unique() in the sense that this function only eliminates consecutive duplicate values. This semantics is similar to std::unique in C++.

input (Tensor) – the input tensor

return_inverse (bool) – Whether to also return the indices for where elements in the original input ended up in the returned unique list.

return_counts (bool) – Whether to also return the counts for each unique element.

dim (int) – the dimension to apply unique. If None, the unique of the flattened input is returned. default: None

A tensor or a tuple of tensors containing output (Tensor): the output list of unique scalar elements.

inverse_indices (Tensor): (optional) if return_inverse is True, there will be an additional returned tensor (same shape as input) representing the indices for where elements in the original input map to in the output; otherwise, this function will only return a single tensor.

counts (Tensor): (optional) if return_counts is True, there will be an additional returned tensor (same shape as output or output.size(dim), if dim was specified) representing the number of occurrences for each unique value or tensor.

(Tensor, Tensor (optional), Tensor (optional))